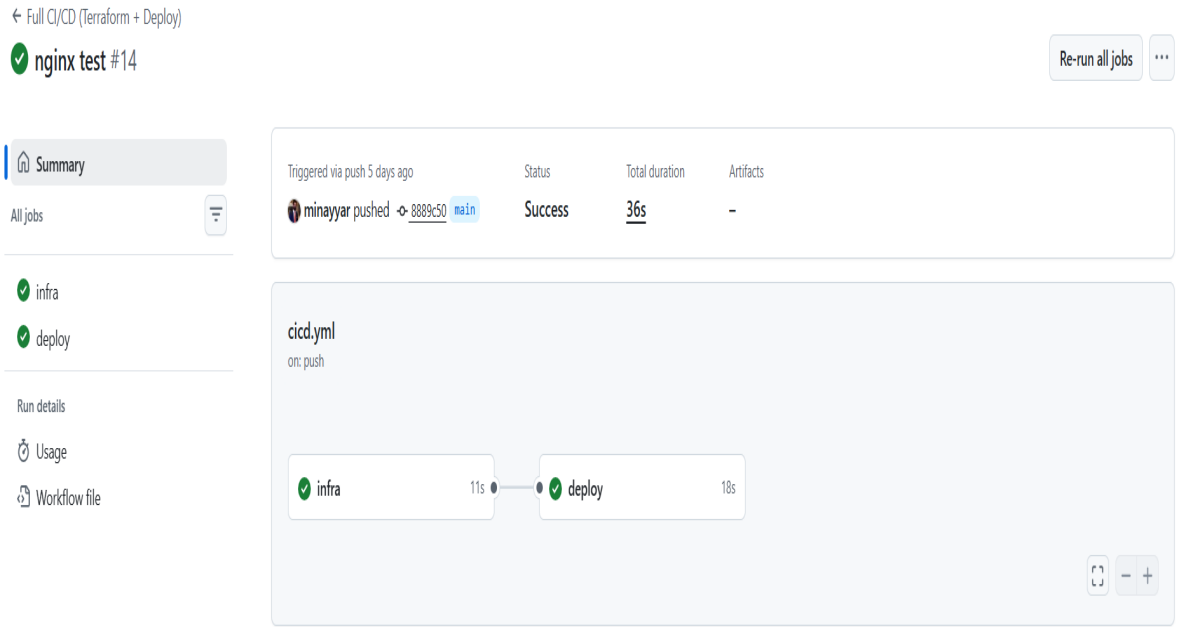
DevOps End-to-End Implementation Report

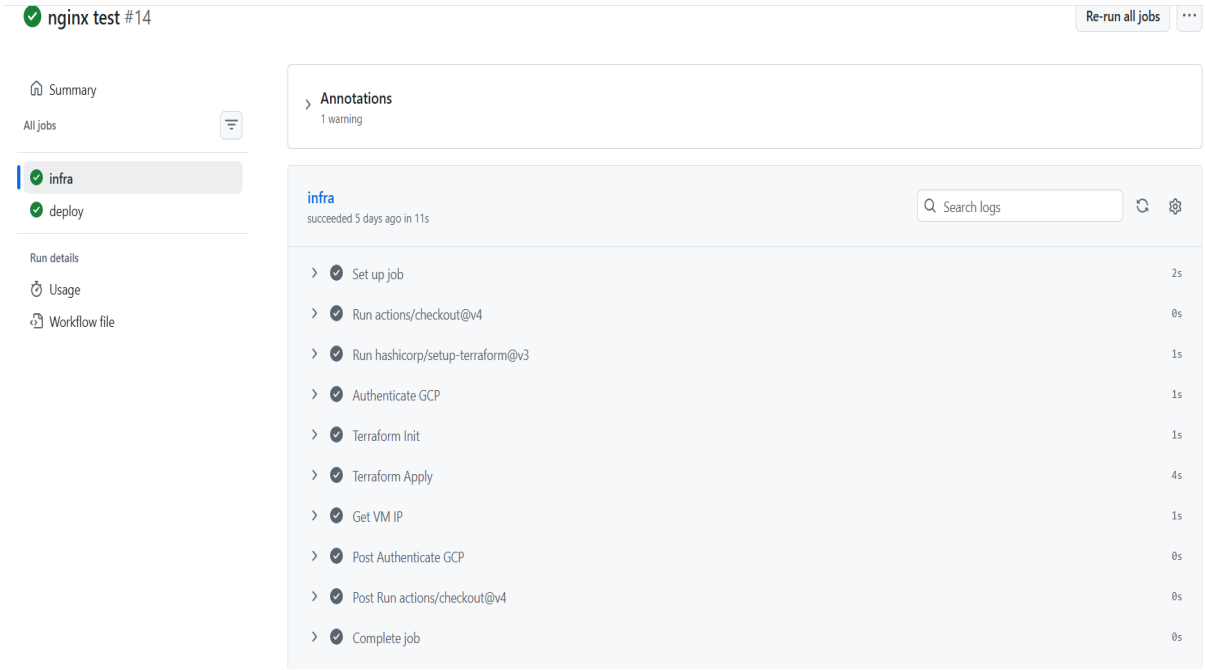
Task 1: Infrastructure Provisioning (Terraform)

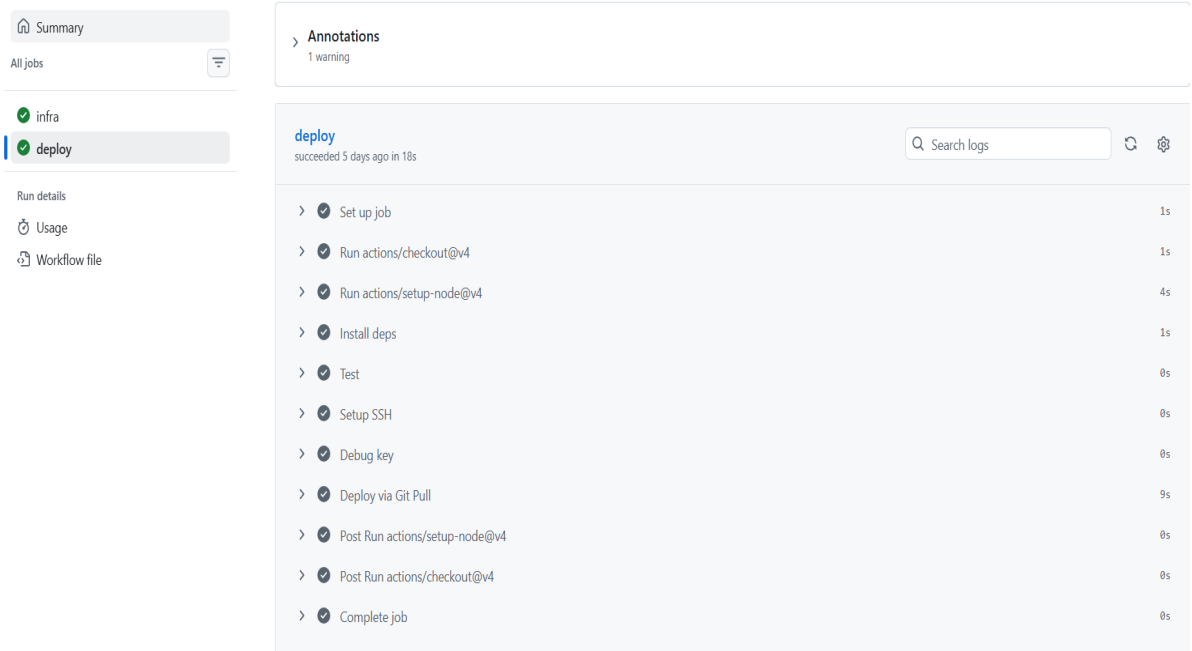
Infrastructure was provisioned on Google Cloud Platform using Terraform. A custom VPC, subnet, firewall rules (HTTP, HTTPS, SSH), and an Ubuntu-based VM were created. Remote state management was implemented using Google Cloud Storage to ensure consistency and collaboration. A startup script was used to install and run a web server.

Task 2: CI/CD Pipeline (GitHub Actions)

A full CI/CD pipeline was implemented using GitHub Actions. The pipeline consists of two jobs: infrastructure provisioning and application deployment. Terraform is executed to provision infrastructure, after which the VM IP is retrieved. The application is then built, tested, and deployed via SSH. PM2 is used to manage the Node.js application, and Nginx is configured as a reverse proxy.

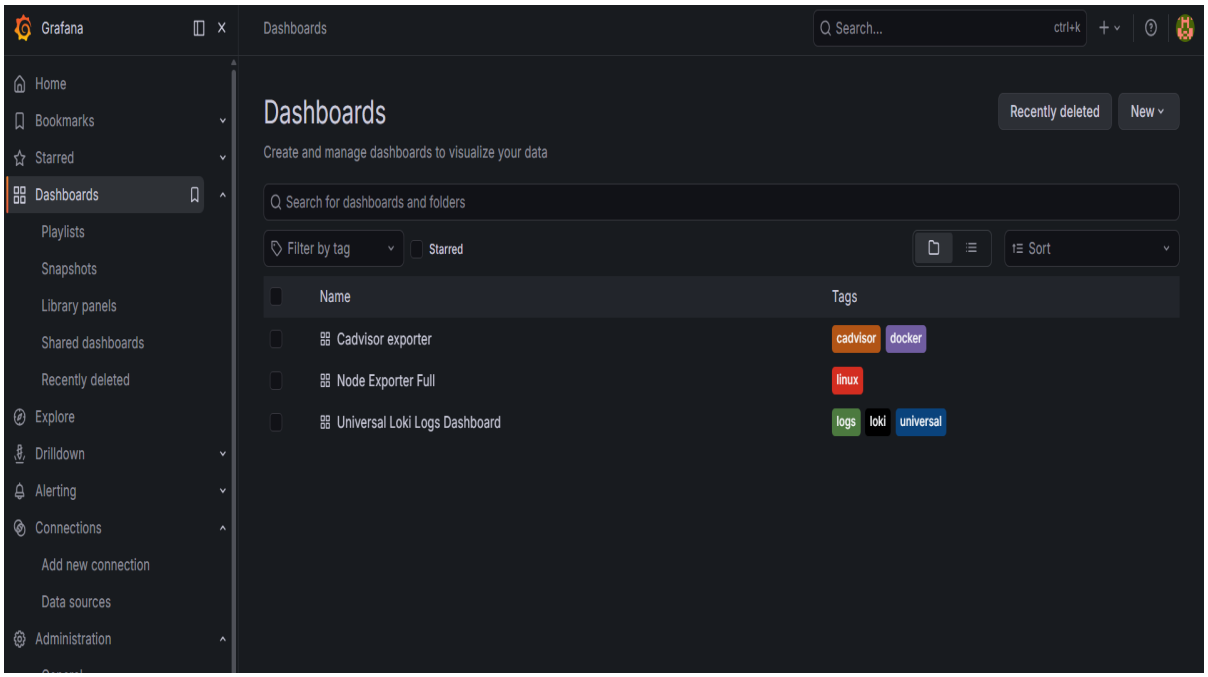






Task 3: Monitoring & Logging

Monitoring and logging were implemented using open-source tools. Prometheus collects metrics from Node Exporter and cAdvisor. Grafana provides dashboards for visualization. Loki and Promtail are used for log aggregation and analysis. This setup enables tracking of system performance, container usage, and application logs in real-time.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Add new connection
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Administration
Dashboards
Recently deleted
New
Create and manage dashboards to visualize your data
Search for dashboards and folders
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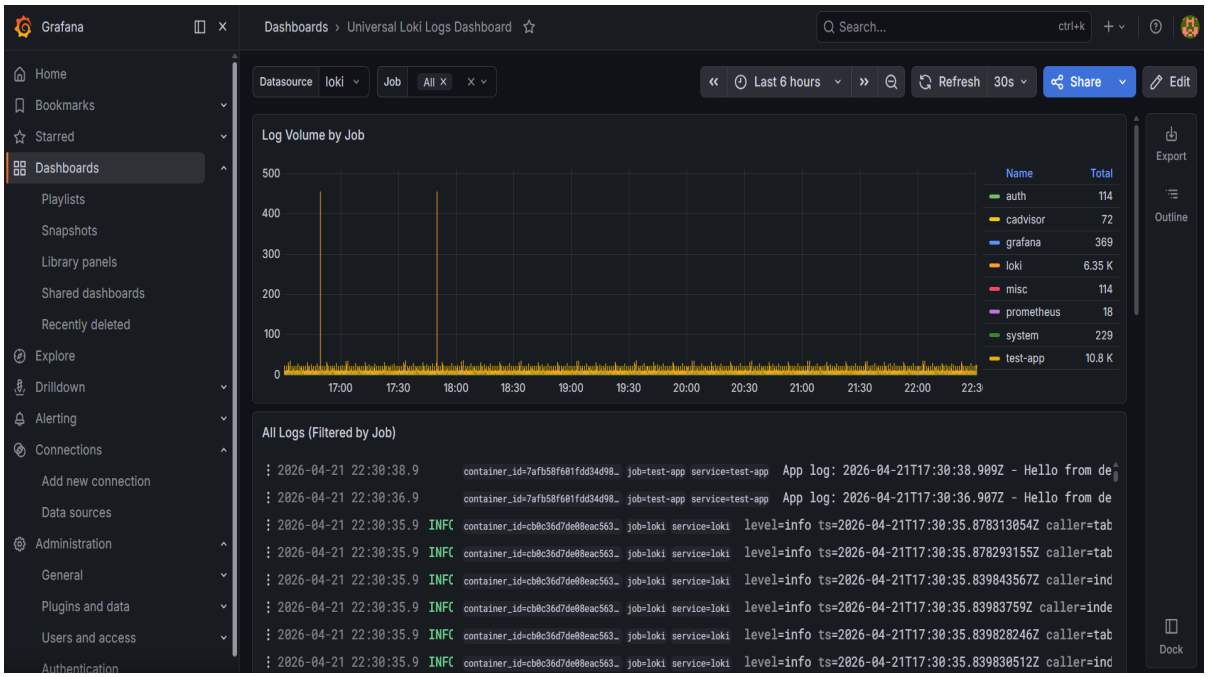
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cadvisor
72
grafana
369
loki
6.35 K
misc
114
prometheus
18
system
229
test-app
10.8 K
Export
Outline
Dock
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App log: 2026-04-21T17:30:38.909Z - Hello from de
2026-04-21 22:30:36.9
App log: 2026-04-21T17:30:36.907Z - Hello from de
2026-04-21 22:30:35.9
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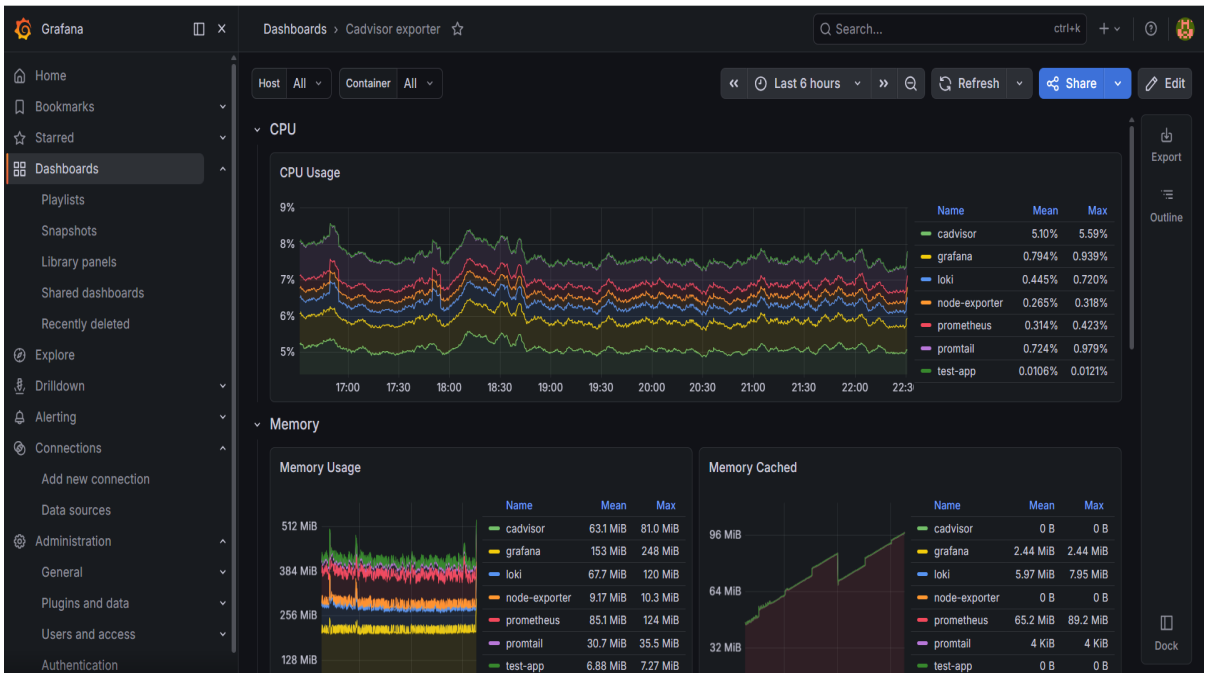

Task 4: Security & Best Practices

Security best practices were applied across the system. Sensitive credentials are managed using GitHub Secrets. SSH access is restricted via firewall rules and secured using key-based authentication. Nginx acts as a reverse proxy to protect backend services. HTTPS can be enabled using Let's Encrypt for secure communication. The principle of least privilege is followed for service accounts.

Challenges & Solutions

SSH authentication issues were resolved by properly formatting and validating private keys. Terraform state inconsistencies were solved using a remote backend. Monitoring setup complexity was handled using Docker Compose for easy orchestration of services.

Conclusion

This project demonstrates a complete DevOps workflow including infrastructure provisioning, CI/CD automation, monitoring, and security. The system is scalable, automated, and follows industry best practices.